

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

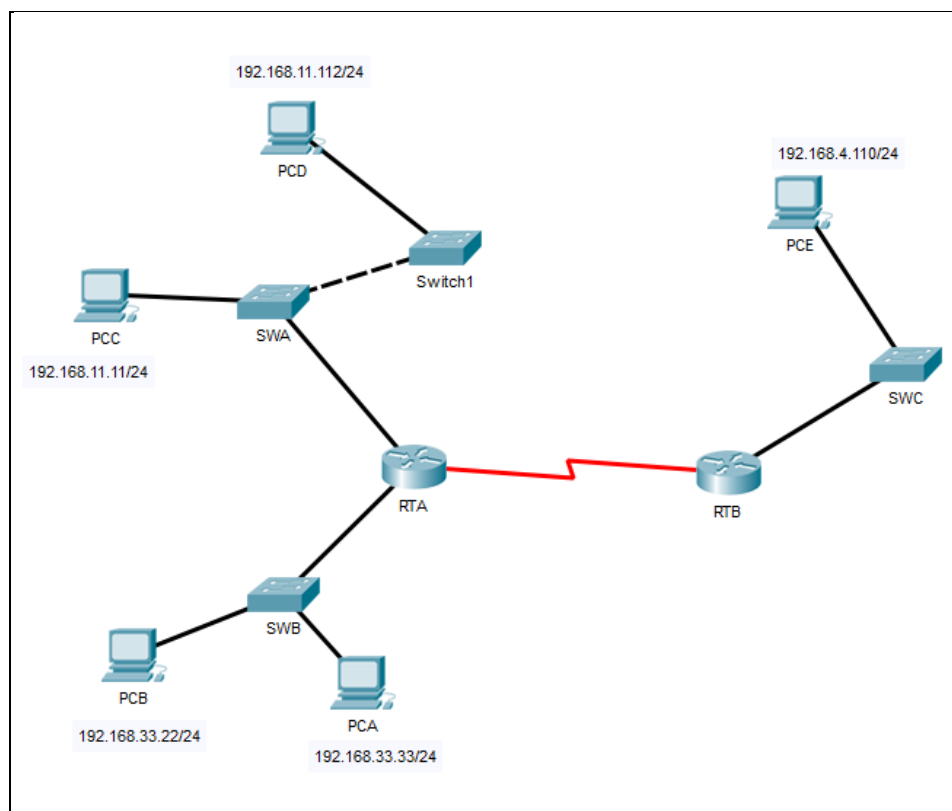


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
RTA#
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
RTB#
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

SWA

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.4a00.0e91    DYNAMIC Fa0/1
1       000c.8546.7d85    DYNAMIC Fa1/1
SWA#
```

SWB

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a    DYNAMIC Fa0/1
SWB#
```

SWC

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614   DYNAMIC Fa0/1
SWC#
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PCA

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCB

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

- e. What are your thoughts on the results?
Ans: There are no entry for all PCs because all the PCs didn't receive any ARP request so there will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- a. Click **PCA** and click the Desktop tab > Command Prompt.
- b. Enter the **ping 192.168.33.22** command. This may take a few seconds.
- c. In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.33.22         0060.47ea.a746       dynamic

C:\>
```

- d. In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.33.33         0002.1755.9a06       dynamic

C:\>|
```

- e. In the Command prompt of PCC, PCD abd PCE, type **arp -a**. Paste the result of this command here.

PCC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCE

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>|
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time=16ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 5ms

C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.11.1          0002.4a00.0e91       dynamic

C:\>|
```

- Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.11.1          0002.4a00.0e91       dynamic
    192.168.11.112        0001.6462.0278       dynamic

C:\>|
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=15ms TTL=126
Reply from 192.168.4.110: bytes=32 time=15ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 15ms, Average = 10ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic

C:\>
```

- e. Discuss the results you got from all the commands on PCC.
Ans: The entries of ARP are shown in the ARP table only after PCC pings PCB, PCC and PCE. There are no ARP entries are found before these ping activities. The ARP table displays the entries once PCB, PCD and PCE receive the ARP request from PCC and subsequently receive pings from PCB< PCD and PCE.
- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address      Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1  -          0002.4A00.0E91 ARPA   FastEthernet1/0
Internet  192.168.11.11 6          00D0.D39A.C0D9 ARPA   FastEthernet1/0
Internet  192.168.33.1  -          000C.CF0C.593A ARPA   FastEthernet0/0
Internet  192.168.33.22 6          0060.47EA.A746 ARPA   FastEthernet0/0
RTA#
```

- g. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
Internet 192.168.4.110 6 0060.702D.7C08 ARPA FastEthernet0/0
RTB#
```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.6462.0278   DYNAMIC   Fa1/1
1       0002.4a00.0e91   DYNAMIC   Fa0/1
1       000c.8546.7d85   DYNAMIC   Fa1/1
1       00d0.d39a.c0d9   DYNAMIC   Fa2/1
SWA#
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a    DYNAMIC Fa0/1
SWB#
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614    DYNAMIC Fa0/1
1       0060.702d.7c08    DYNAMIC Fa1/1
SWC#
```

- d. Do switches use arp table? (Y/N)
Ans: Yes
- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*
Ans: Layer 3 switches use an Address Resolution Protocol (ARP) table to map IP addresses to MAC addresses. ARP tables are used to forward packets or route them based on a routing protocol.
- f. What information does the command **show mac-address-table** gives?
Ans: To view the MAC address table entries on a switch, the term 'DYNAMIC' specifically pertains to display MAC address that have been learned by the switch during its operation.

Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- Change the Lab 4 filename to include your name. *Example: Lab4AliAhmad.pkt
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot shows the Cisco Packet Tracer interface. On the left, a network topology is displayed with a switch (S1) connected to three PCs (PC1, PC2, PC3) and a wireless router (WRS2). The switch has three subinterfaces: G0/0.10 (172.17.10.1/24), G0/0.20 (172.17.20.1/24), and G0/0.88 (172.17.88.1/24). PC1 is connected to S1 G0/0.10, PC2 to S1 G0/0.20, and PC3 to WRS2. WRS2 is connected to S1 G0/0.88. The PT Activity window on the right shows the 'Configuring Wireless LAN Access' activity. It includes an 'Addressing Table' and a 'Completion' status of 100/100.

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

Completion: 100/100

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

The screenshot shows the Cisco Packet Tracer interface with the 'Part 2: Configure a Wireless Client' activity. The activity is divided into three parts: Part 1 (Configure PC3 for wireless connectivity), Part 2 (Verify PC3 wireless connectivity and IP addressing configuration), and Part 3 (Verify Connectivity). The completion status is 100/100.

Part 2: Configure a Wireless Client

Step 1: Configure PC3 for wireless connectivity.

Because SSID broadcast is disabled, you must manually configure PC3 with the correct SSID and passphrase to establish a connection with the router.

- Click PC3 > Desktop > PC Wireless.
- Click the Profiles tab.
- Click New.
- Name the new profile Wireless Access.
- On the next screen, click Advanced Setup. Then manually enter the SSID of WRS_LAN on Wireless Network Name. Click Next.
- Choose Obtain network settings automatically (DHCP) as the network settings, and then click Next.
- On Wireless Security, choose WPA2-Personal as the method of encryption and click Next.
- Enter the passphrase cisco123 and click Next.
- Click Save and then click Connect to Network.

Step 2: Verify PC3 wireless connectivity and IP addressing configuration.

- The Signal Strength and Link Quality indicators should show that you have a strong signal.
- Click More Information to see details of the connection including IP addressing information.
- Close the PC Wireless configuration window.

Part 3: Verify Connectivity

All the PCs should have connectivity with one another.

Completion: 100/100

Cisco Packet Tracer - C:\Users\Evelyn Goh\Desktop\Lab 4 Evelyn Goh.pka - Guest - 2025-01-17 10:18:45

File Edit Options View Tools Extensions Window Help

Activity Results Time Elapsed: 00:18:15

Congratulations Guest! You completed the activity.

[Overall Feedback](#) [Assessment Items](#) [Connectivity Tests](#)

Completed Feedback: Congratulations! You successfully completed the **Configuring Wireless LAN Access** activity. However, your final score may change based on your answers to the questions in the Instructions. Consult your instructor.

Cisco Packet Tracer - C:\Users\Evelyn Goh\Desktop\Lab 4 Evelyn Goh.pka - Guest - 2025-01-17 10:18:45

File Edit Options View Tools Extensions Window Help

Activity Results Time Elapsed: 00:18:51

Congratulations Guest! You completed the activity.

[Overall Feedback](#) [Assessment Items](#) [Connectivity Tests](#)

[Expand/Collapse All](#) [Show Incorrect Items](#)

Assessment Items	Status	Points	Component(s)	Feedback
Network				
PC3				
Wireless				
Security Mode				
Authn Type	Correct	1	Wireless Client ...	
Pass Phrase	Correct	4	Wireless Client ...	
SSID	Correct	5	Wireless Client ...	
WRS2				
(deprecated) DHCP Server				
(deprecated) DHCP Enable	Correct	10	Wireless Router ...	
(deprecated) Pools		0	Ip	
(deprecated) Pool linksysPool		0	Ip	
(deprecated) Default Gateway	Correct	10	Wireless Router ...	
Default Gateway	Correct	10	Wireless Router ...	
Ports				
Internet				
IP Address	Correct	10	Wireless Router ...	
Link to S1				
Connects to FastEthernet0/7	Correct	5	Device Connection	
Type	Correct	5	Device Connection	
Wireless		0	Other	
Wireless				
Security Mode				
Authn Type	Correct	10	Wireless Router ...	
Pass Phrase	Correct	10	Wireless Router ...	
SSID	Correct	10	Wireless Router ...	
SSID Broadcast	Correct	10	Wireless Router ...	

Score : 100/100

Item Count : 13/13

Component	Items/Total	Score
Device Connection	2/2	10/10
Wireless Client Configuration	3/3	10/10
Wireless Router Configuration	8/8	80/80